



Performance of a plate-fin heat exchanger with vortex-generator channels: 3D-CFD simulation and experimental validation



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ABSTRACT

A type of plate-fin channel with rectangular wings as a transverse vortex-generator channel was considered. The heat transfer and fluid flow characteristics of the vortex-generator channel were systematically analyzed by 3D-CFD simulations. As the first part of the project, a precise and reliable experimental setup was designed and fabricated to generate a constant temperature boundary condition. Inlet–outlet bulk and surface temperatures along with flow rate data were acquired by using pt-100 and K series calibrated thermocouples and an ultrasonic flow meter, respectively. Hence, boundary conditions for the CFD simulations were well defined, and data for the validation of a reference model were generated. Comparing the experimental data with the predicted results by CFD simulation implies that the applied CFD approach can properly predict the thermal-hydraulic specifications of the vortex-generator channel. In the second part of the project, the influences of seven effective geometrical parameters (i.e. wings height, wings width, channel length, longitudinal wings pitch, transverse wings pitch, wings attack angle, and wings attack angle) for three conventional coolants flow (i.e. water, oil, and ethylene glycol) at the laminar flow regime were evaluated. As the third part of the project, general correlations were derived for Nusselt number of the coolants flow inside the plate-fin heat exchangers with vortex-generator channels.

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1. Introduction

Plate-fin heat exchangers (PFHEs) are widely used in many industrial areas such as chemical and food engineering, radiator of car, power system, heating and air conditioning, refrigeration, aerospace, and other engineering applications. Hence, the high efficiency, small size, light weight, and low cost are the desirable and significant criteria in the fabrication of these types of heat exchanger. Various types of the plate-fin channels such as plain, perforated, offset strip, louvered, wavy, vortex-generator, and pin have been used in the PFHEs. Comparative studies [1–3] depict that the vortex-generator channel is the promising surface to attain the mentioned desirable criteria in the PFHEs. Vortex-generators in a channel not only disturb the flow field but also make the fluid swirling and cause a heavy exchange between the core and the wall fluids [4]. The vortex-generators may be divided into the transverse and longitudinal types according to their

rotating axes directions. The axes of transverse and longitudinal vortex-generators lie perpendicular and parallel to the main flow direction, respectively.

Biswas et al. reported that the longitudinal vortex-generators show a great promise for enhancing the heat transfer rate in plate-fin heat exchangers [5,6] and fin-tube heat exchangers [7,8]. They concluded that the presence of a winglet type longitudinal vortex-generator in the wake region behind the cylinder increases the heat transfer in this region as high as 240% [8]. Appropriate thermal-hydraulic specifications of the vortex-generators have also motivated many other investigators to study and optimize the performance of these geometries in different heat exchange systems such as circular and non-circular ducts [9–14], plate-fin heat exchanger [15–19], tube-fin heat exchanger [20–28], heat sink [29,30], electronic chip [31,32], and micro-and-mini channels [33]. A comprehensive review of studies on the vortex-generator applications has been recently conducted by Ahmed et al. [34]. The literature shows that the studies on heat transfer characteristics of vortex-generators inside the PFHEs are scarce. Gentry and Jacobi [15] studied the effect of delta-wing vortex-generator embedded in a flat plate. They found that the heat transfer was enhanced up to 50% compared to the plain flat plate. Kotcioğlu et al. [16] investigated the thermal performance in a plate fins

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channel using the four basic types of vortex-generators. They pointed out that the mixing effect between wing cascades improves the heat transfer characteristics, as the inclination angle of winglet increases. Chen and Shu [17] showed that the presence of delta-wing vortex-generator mounted on heated plate installed on the bottom wall of a duct has little overall effect on the near-wall averaged axial mean velocity and axial vorticity, but increases the turbulent kinetic energy. Wu and Tao [18] made a study to investigate the heat transfer characteristics of a plain plate and four plates with a pair of delta-winglet longitudinal vortex-generator with attack angles ranged from 15° to 60° . They reported that the average Nusselt number of plate with attack angle of 60° is slightly higher than that of plate with attack angle of 45° yet may bring larger pressure drop. Akcayoglu [19] performed an investigation about the heat transfer enhancement in a PFHE with triangular fins as insert using a winglet pair of vortex-generator. It was confirmed that the average Nusselt number increases with an increase in the angle of attack and Reynolds number.

In the noted literature, there are few studies in detailed analysis of heat transfer enhancement by the transverse vortex-generator PFHEs, when liquids are applied as working media. Also, a parametric study on the transverse vortex-generator channels with rectangular wings has been not performed in the earlier works. In the present study, an experimental study of water flow in a transverse vortex-generator channel with rectangular wings was carried out. In addition to the experiment, three-dimensional (3D) numerical simulations (Computational Fluid Dynamics – CFD) were carried out to provide a more comprehensive quantitative and qualitative assessment which was not easily accessible in the experimental study [35]. The first objective is to validate the CFD results with the obtained experimental data. The second one is to evaluate the influence of wings shape on the performance of the considered vortex-generator channel. The third one is to study the effects of seven geometrical parameters on the fluid flow and heat transfer specifications of the transverse vortex-generator channels with rectangular wings. A criterion will be used to evaluate the heat transfer and pressure drop performance of the PFHE. The fourth one is to propose Nusselt number correlations for the studied vortex-generator channels, based on the studied geometrical parameters and considered working fluids.

2. Experimental implementation

2.1. Reference vortex-generator channel

Initially, experiments were carried out for the water flow inside a reference vortex-generator channel to produce required data for boundary conditions definition and results validation of the CFD simulations. Delta wing, rectangular wing, delta winglet, and rectangular winglet are the four basic shapes of the vortex-generators [36]. These tiny protrusions may be incorporated into the main surface by punching, stamping, attaching, mounting, or embossing [37]. In the present study, the reference channel with rectangular wings was generated from the main plates by using the punching technique. The channel consists of three wings in each row; one mounted to the top plate and the other ones mounted to the bottom plate. This trend alternately changes between the top and the bottom plates. The configuration of vortex-generator channels with rectangular wings is described by the geometrical parameters, including wing height (w_h), wing width (w_w), channel length (L), longitudinal wings pitch (l_p), transverse wings pitch (t_p), wings attach angle (α), and wings attack angle (β). The corresponding values of the reference channel were 5.0 mm,

400 mm, 20 mm, 10 mm, 90° , and 90° , respectively. The experimental test section was composed of the vortex-generator channel, two cover plates, two side bars, a header, and a nozzle. The reference model and experimental test section are displayed in Fig. 1.

2.2. Experimental setup

A schematic diagram of the designed experimental test loop is shown in Fig. 2. The main sections of experimental apparatus are (1) transmission fluid state, (2) measurement equipment, (3) constant temperature bath system, and (4) cooling unit. To achieve reliable and accurate results for the validation of CFD simulations, very sensitive measurement equipment and also precise control systems were utilized. For instance, the volumetric flow rate passing through the test section was measured by using a high sensitive ultrasonic flow meter (Flownetix® 100series™). This instrument, which its accuracy was estimated to be $\pm 0.05 \text{ lit min}^{-1}$, utilized of sound speed information gained from ultrasonic signals to automatically correct for any changes in fluid physical properties. The pressure drop was found by subtracting the measured local pressure values at the inlet and outlet of the test section. To achieve precise results, two very sensitive pressure transmitters (PSCH0.05BCIA, Sensys) with accuracy of $\pm 1.0 \text{ Pa}$ were utilized. Two pt-100 thermocouples were used to measure the inlet and outlet temperatures of the fluid by putting them into the flow line. Six K-type thermocouples were mounted on the external and different positions of the test sections surfaces to measure the wall temperature distribution.

2.3. Experimental uncertainties

The uncertainty of Colburn factor (j) and Fanning friction factor (f) were conducted implementing the error values of instruments in Section 2.2 based on the method proposed by Kline and McClintock [38]. The uncertainty values of Colburn factor, Fanning friction factor, and Reynolds number were calculated about 2.19%, 3.27%, and 1.82%.

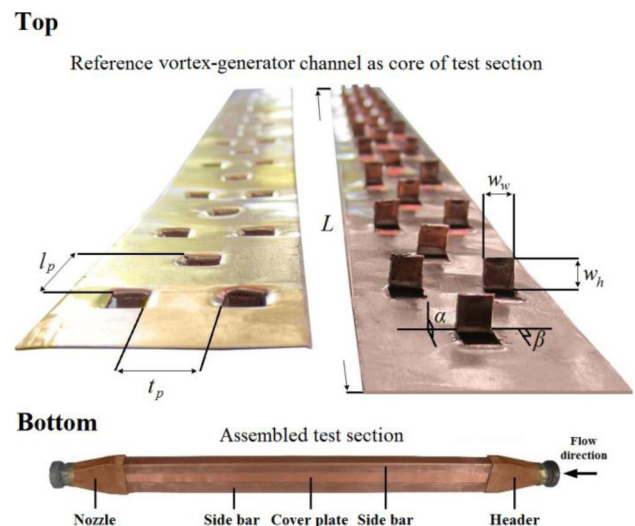


Fig. 1. Reference vortex-generator channel (top) and assembled test section (bottom).

Table 1
Geometrical parameters for studied models.

Parameter	Symbol	Value
Wing height	w_h (mm)	2.5, 5.0, 7.5
Wing width	w_w (mm)	2.5, 5.0, 7.5
Channel length	L (mm)	300, 400, 500
Longitudinal wings pitch	l_p (mm)	15, 20, 25
Transverse wings pitch	t_p (mm)	5.0, 7.5, 10
Wings attach angle	α (°)	45, 60, 90
Wings attack angle	β (°)	45, 60, 90

$$u_x = u_y = u_z = 0, \quad \frac{\partial T}{\partial y} = 0 \quad (1c)$$

(b) In the middle region (core part of the computational domain):

At the lower and upper boundaries (2-3-6-7 and 10-11-14-15)

$$u_x = u_y = u_z = 0, \quad T = T_{\text{wall}} = \text{const} \quad (2a)$$

At the left and right boundaries (2-3-11-10 and 7-6-14-15)

$$u_x = 0, \quad \frac{\partial u_y}{\partial x} = \frac{\partial u_z}{\partial x} = 0, \quad \frac{\partial T}{\partial x} = 0 \quad (2b)$$

(c) In the downstream-extended region (outlet fluid block):

At the left and right boundaries (3-4-12-11 and 6-5-13-14)

$$u_x = u_y = u_z = 0, \quad \frac{\partial T}{\partial x} = 0 \quad (3a)$$

At the lower and upper boundaries (3-4-5-6 and 11-12-13-14)

$$u_x = u_y = u_z = 0, \quad \frac{\partial T}{\partial y} = 0 \quad (3b)$$

At the outlet boundary (4-12-13-5)

$$\frac{\partial u_x}{\partial z} = \frac{\partial u_y}{\partial z} = \frac{\partial u_z}{\partial z} = 0, \quad \frac{\partial T}{\partial z} = 0 \quad (3c)$$

3.3. Numerical method and governing equations

A fluid-solid conjugated simulation was carried out to attain validly imitative practical conditions. In this 3-D simulation the following assumptions were made.

- Newtonian fluids in single phase and incompressible,
- Continuous, steady state, and laminar flow,
- Negligible viscous dissipation, radiation, and natural heat transfer,

Under the above assumptions, the governing equations for the mathematical model were as follows.

- Conservation of mass for fluid

$$\frac{\partial}{\partial x_i} (\rho u_i) = 0 \quad (4)$$

- Conservation of momentum for fluid

$$\frac{\partial}{\partial x_i} (\rho u_i u_k) = \frac{\partial}{\partial x_i} \left(\mu \frac{\partial u_k}{\partial x_i} \right) - \frac{\partial p}{\partial x_k} \quad (5)$$

- Conservation of energy for fluid

$$\frac{\partial}{\partial x_i} (\rho u_i T) = \frac{\partial}{\partial x_i} \left(\frac{\kappa_f}{C_p} \frac{\partial T}{\partial x_i} \right) \quad (6)$$

- Conservation of energy for solid

$$\frac{\partial}{\partial x_i} \left(\kappa_s \frac{\partial T}{\partial x_i} \right) = 0 \quad (7)$$

It should be noted that the material of solid phase was copper with the constant properties. Also, the height of cover plates and the thickness of vortex-generator channels were 0.7 mm and 0.4 mm, respectively.

The governing equations were discretized by the finite volume method (FVM). The standard scheme was used for the pressure discretization, and the semi implicit method for pressure linked equation (SIMPLE) algorithm was applied to deal with the coupling of pressure and velocity. The diffusion term in the momentum and energy equations was approximated by second-order central difference which gives a stable solution. In addition, a second-order upwind scheme was adopted for the discretization of convection terms. The aforesaid governing equations were solved by a segregated implicit iterative scheme. To ensure the computational convergence, the under-relaxation factors were 0.3, 0.7, and 0.8 for the pressure, momentum, and energy, respectively. Before proceeding with computations, the appropriate convergence criteria were obtained. It was done by taking different convergence criteria and checking the difference. The solutions were compared for convergence criteria between 10^{-4} and 10^{-6} for the continuity and momentum equations and 10^{-6} and 10^{-8} for the energy equation. It was found that the difference in results for the heat transfer coefficient values of some models in the distinct conditions was below 0.06%. Hence, the convergence criteria for the continuity, momentum, and energy equations were selected equal to 10^{-4} , 10^{-4} , and 10^{-6} , respectively. The governing equations were iteratively solved until the set residuals were obtained or when they stabilized at constant values. Most numerical models satisfy these criteria by taking the flow to be laminar [39].

4. Grid dependence check and validation procedure

Grid dependency of the result and validation of the studied model are discussed in this section. In compact surface terminologies, the heat transfer coefficient and pressure drop are usually expressed in terms of dimensionless factors. Two well-known dimensionless parameters, Colburn factor, j , and Fanning friction factor, f , were used to describe the heat transfer performance and pressure drop characteristics, Eqs. (8) and (9).

$$j = \frac{h}{\rho u C_p} \text{Pr}^{2/3} \quad (8)$$

$$f = \frac{2\Delta P}{\rho u^2} \frac{D_h}{L_d} \quad (9)$$

Table 2
Grid dependency investigation.

Test no.	Mesh no. in x-direction	Mesh no. in y-direction	Mesh no. in z-direction	Total mesh no.	j Factor value	j_{i+1} & j_i deviation (%)	f Factor value	f_{i+1} & f_i deviation (%)
1	11	25	909	259 200	0.0124	—	0.3218	—
2	13	28	1000	336 340	0.0143	15.55	0.3132	4.43
3	14	31	1143	471 240	0.0163	13.45	0.3009	3.93
4	16	35	1280	737 280	0.0183	12.13	0.2871	4.60
5	18	39	1422	1 035 360	0.0191	4.75	0.2852	0.43
6	20	44	1587	1 407 120	0.0199	3.84	0.2847	0.39
7	22	48	1738	1 837 440	0.0200	0.78	0.2841	0.21

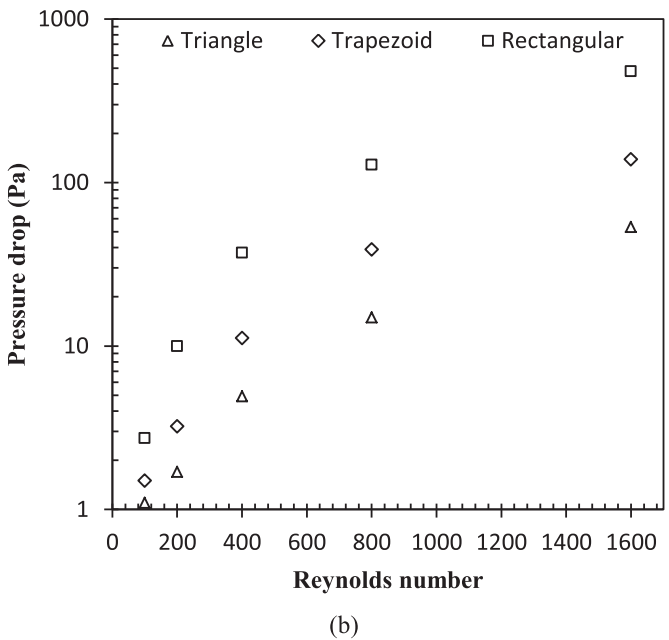
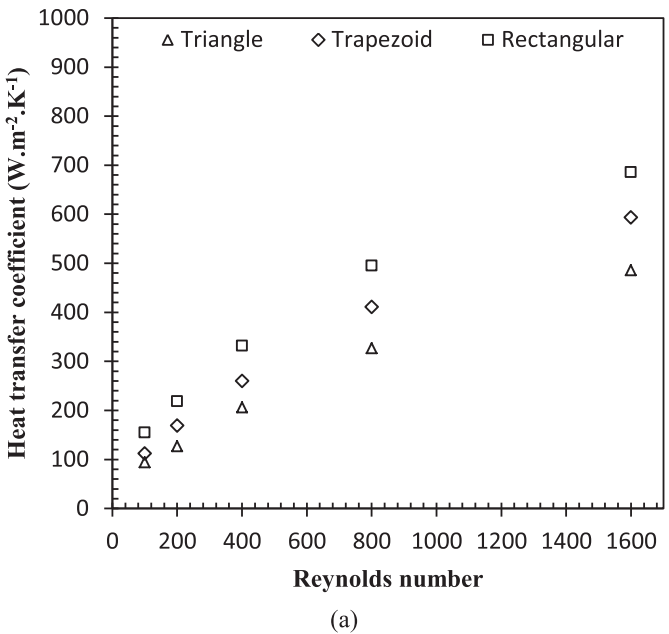
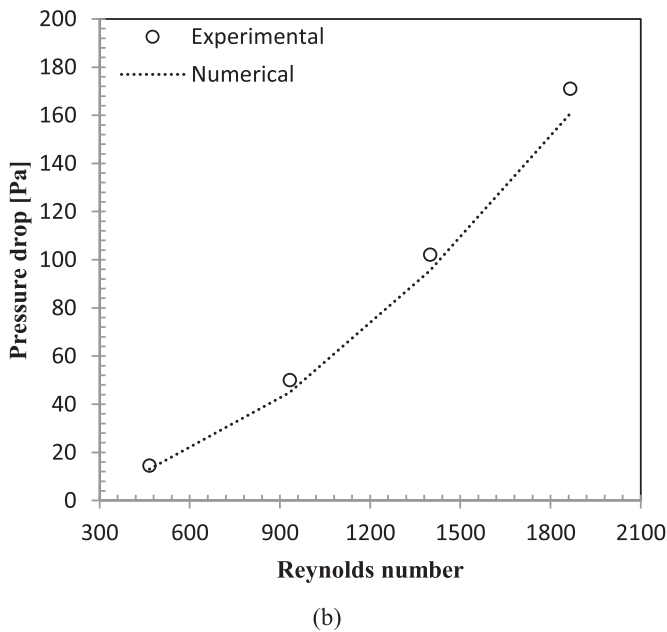
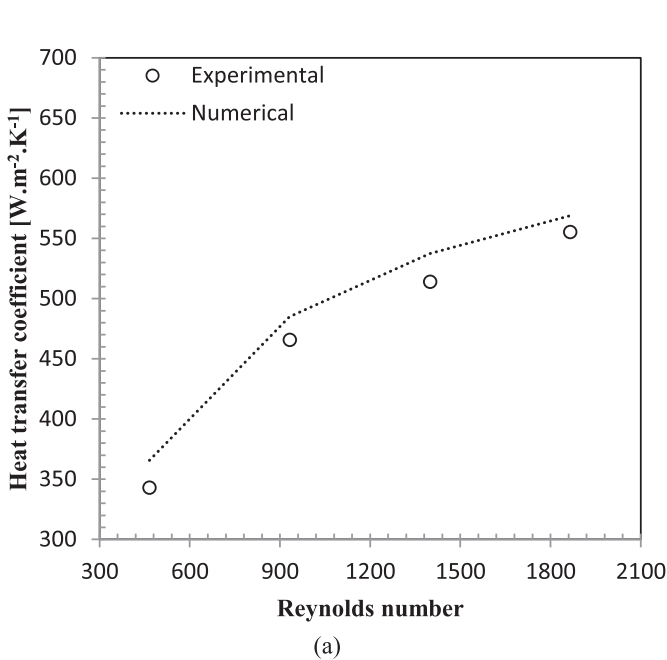


Fig. 4. Comparison of (a) heat transfer coefficient and (b) pressure drop values between computed CFD results and recorded experimental data.

Fig. 5. (a) Heat transfer coefficient – Reynolds number (b) pressure drop – Reynolds number of water flow inside vortex-generator channel with different wings shape.

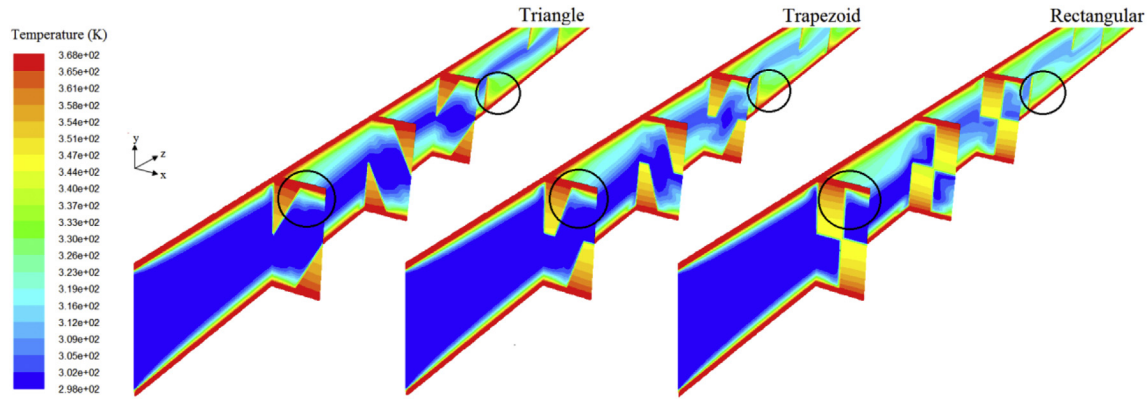


Fig. 6. Temperature distribution inside vortex-generator channels with different wing shapes.

where, ρ is the fluid density, u is the mean velocity of the minimum transverse area, C_p is the specific heat of working fluids, Pr is the Prandtl number, D_h is the hydraulic diameter, and L_d is the channel length, and h is the convective heat transfer coefficient,

$$h = \frac{Q_{\text{conv}}}{A_t(T_w - T_b)_{\text{LMTD}}} \quad (10)$$

where, Q_{conv} is the convective heat transfer rate which the fluid flowing through the test section absorbs from the hot surfaces. The absorbed heat by the fluid can be expressed, as follows,

$$Q_{\text{conv}} = mC_p(T_{b,\text{out}} - T_{b,\text{in}}) \quad (11)$$

where, m is the mass flow rate, and $T_{b,\text{in}}$ and $T_{b,\text{out}}$ are the average temperature of fluid at the inlet and outlet of test section, respectively. Also in Eq. (10), A_t is the total heat transfer area, and $(T_w - T_b)_{\text{LMTD}}$ is the log mean temperature difference, as follows,

$$(T_w - T_b)_{\text{LMTD}} = \frac{\Delta T_{w-b,\text{in}} - \Delta T_{w-b,\text{out}}}{\log(\Delta T_{w-b,\text{in}} / \Delta T_{w-b,\text{out}})} \quad (12)$$

where, $\Delta T_{w-b,\text{in}}$ and $\Delta T_{w-b,\text{out}}$ denote the differences between the wall temperature and the bulk fluid temperature at the inlet and outlet of heat transfer section.

The ΔP is the pressure difference between inlet and outlet as.

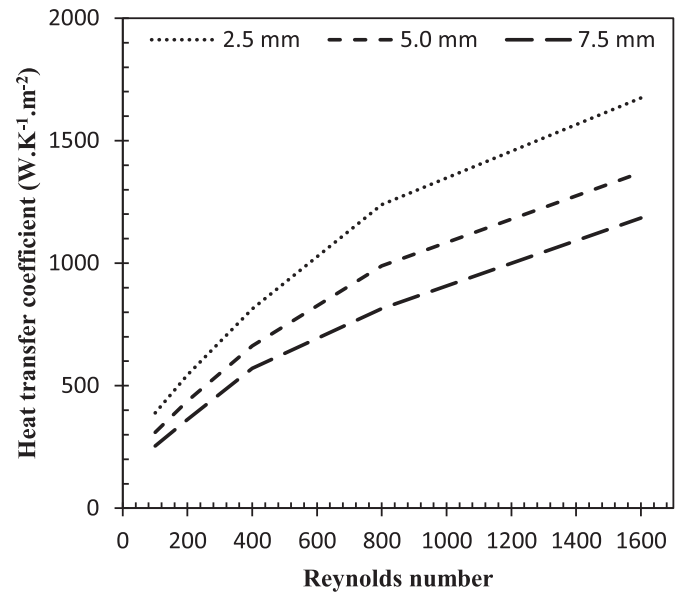
$$\Delta P = P_{\text{av},\text{in}} - P_{\text{av},\text{out}} \quad (13)$$

where, subscripts 'av,in' and 'av,out' refer to the average pressure values at the inlet and outlet cross sections.

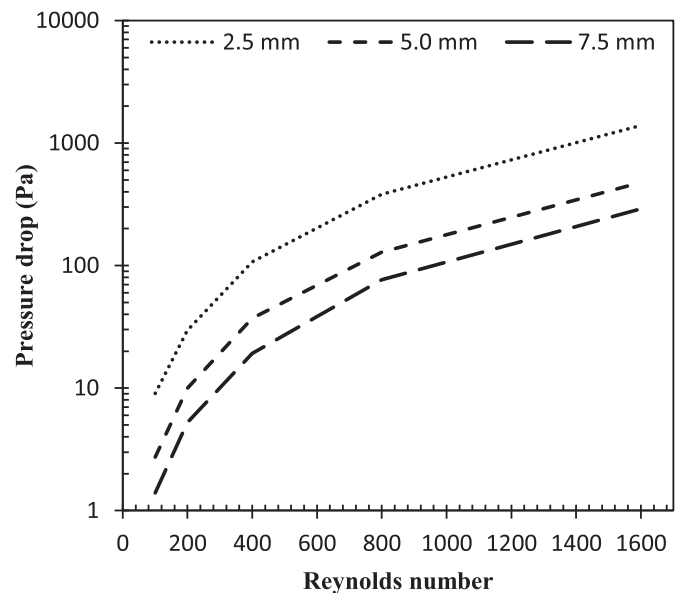
4.1. Grid dependency

The quality and quantity of grids are very important for the accuracy and time of CFD simulations. Therefore, care must be taken in selecting the mesh type and size, so not to affect the CFD results. The structured hexahedral meshes due to the advantages such as arranging in a line with the main flow direction and providing more precise flow solution than the tetrahedral or polyhedral ones were chosen in the current study. It was difficult to use a single structured hexahedral mesh on the whole computational domain particularly for the models with variable attach and attack angles (i.e. 45° and 60°). Therefore, in order to optimize the mesh quality and quantity, the models were accurately divided into simple zones.

The grid dependence was investigated by conducting simulation in a vortex-generator channel with the highest values of different geometrical parameters ($w_h = w_w = 7.5$ mm, $L = 500$ mm,



(a)



(b)

Fig. 7. (a) Heat transfer coefficient – Reynolds number (b) pressure drop – Reynolds number for channels with different wings height.

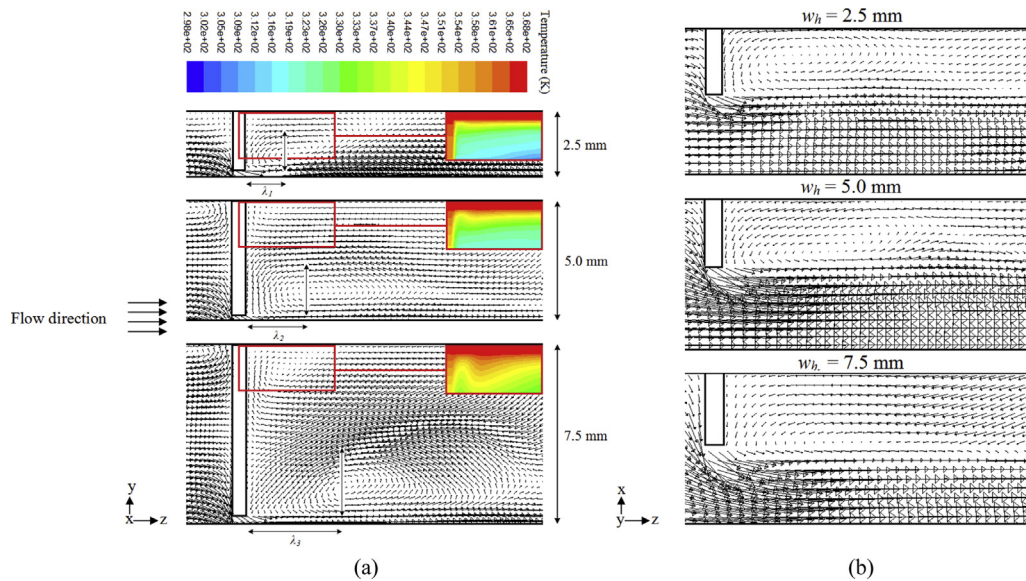


Fig. 8. Effect of wings height variation on (a) transverse swirl flows and (b) longitudinal swirl flows and temperature gradient near wings at $Re = 800$.

$l_p = 25$ mm, $t_p = 10$ mm, and $\alpha = \beta = 90^\circ$) at Reynolds number equal to 1200. Firstly, the variations of j factor and f factor with the degree of grid refinement near the hot walls were checked employing three grid refinement ratios of 1.0, 1.05, and 1.10. The results show that increasing the grid refinement ratio decreases both j factor and f factor values, and these parameters nearly remain at the constant values, when the grid refinement ratio is higher than 1.05. Then, seven different mesh densities about 259 200, 336 340, 471 240, 737 280, 1 035 360, 1 407 120, and 1 837 440 cells were chosen. The variations of predicted j factor and f factor with mesh number are shown in Table 2. It can be seen that both the j factor and the f factor values nearly remain at the constant values, when the grid number is higher than 1 407 120. The number of meshes was increased more than the highest value (i.e. 1 837 440) to show the sufficiency of mesh resolution. It was confirmed that the results were invariable. Thus, the accepted meshes number in the computational domain for all the models was 20 cells in the channel width (x -direction), 44 cells in the channel depth (y -direction), and 1587 cells in the channel length (z -direction), totally 1 407 120 cells. Therefore, the type of grid was non-uniform structured and had a suitable number with appropriate expansion ratio to concentrate the grid highly near the wall.

4.2. Validation procedure

Reynolds number is based on the velocity in the minimum cross sectional flow area and hydraulic diameter.

$$Re = \frac{u_{in} D_h \rho}{\mu} \quad (14)$$

$$D_h = \frac{4LA_c}{A_t} \quad (15)$$

where, L is the channel length, A_c is the minimum cross sectional flow area, and A_t is the total heat transfer surface area. In this study, Reynolds number values are from 200 to 1600. Therefore, the inlet velocities of each model were changed based on its geometrical parameters at different Reynolds numbers.

In order to validate the CFD study, a simulation was carried out at the same geometry, dimensions, and operating conditions, as

presented in the current experimental study. To achieve accurate results, the generated model contains all the experimental test section components (see assembled test section part in Fig. 1). As depicted in Fig. 4, the obtained CFD results show a good agreement with the recorded experimental data. It can be observed that the numerically predicted heat transfer coefficient is about 4.6% higher than that of the experimental data. This is may be attributed to the existence of conduction resistance in the assembled experimental test section, which was not considered in the generated model. Also, the pressure drop of the CFD results is about 8.9% lower than that of the experimental data. Probably, the entrance effects in the real test section such as the thickness of channel and cover plates make the pressure drop of the experimental study higher than the obtained numerical one. The excellent agreement between the numerical results and the experimental data indicates the reliability of present numerical procedure.

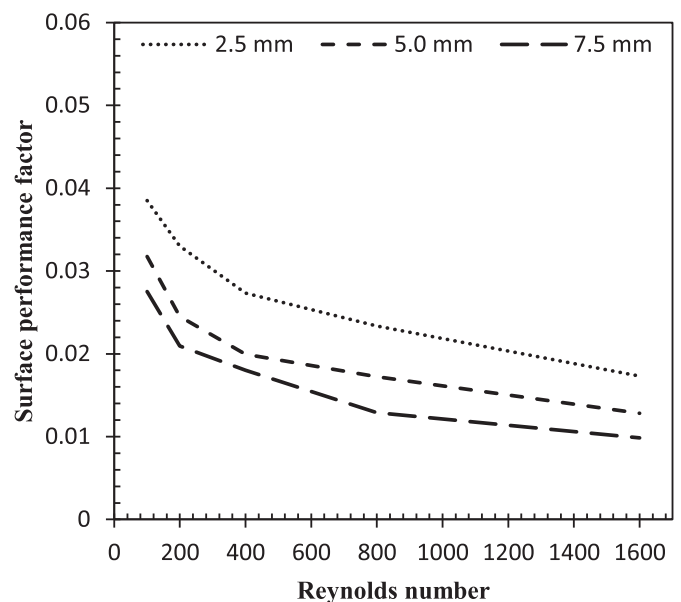
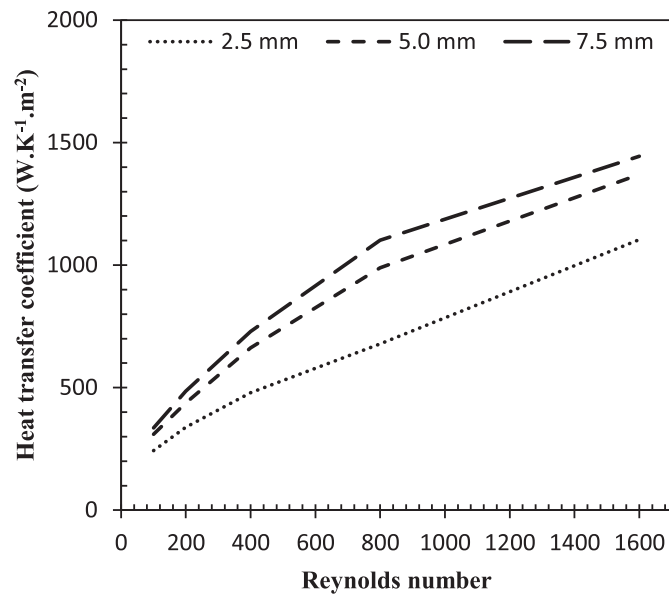


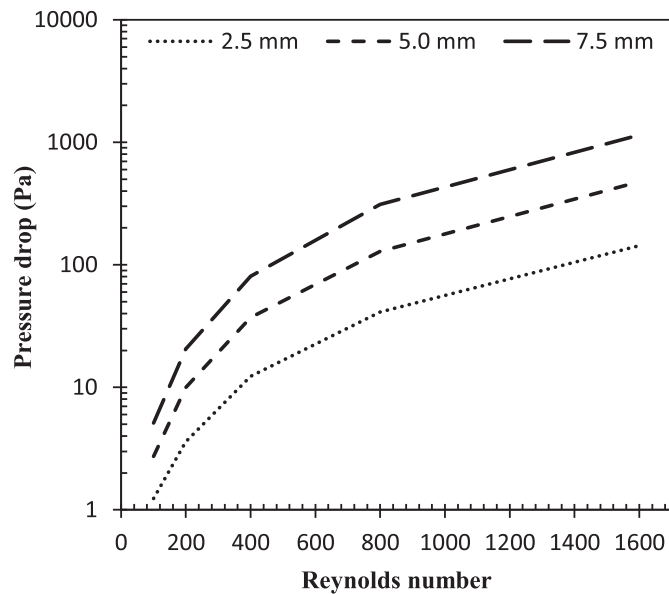
Fig. 9. Surface performance factor – Reynolds number for channels with different wing heights.

5. Results and analysis

The present computations were performed for a 3-D laminar flow of common liquid working fluids, namely water ($Pr = 7$), oil ($Pr = 50$), and ethylene glycol ($Pr = 150$), over the vortex-generator channels. As presented in Table 1, three different values of seven geometrical parameters were considered. Various forms of quantitative and qualitative results can be displayed from the output of simulations, but due to the space restriction only some of these results are presented. The performance is presented in terms of the heat transfer coefficient and pressure drop, and the optimal geometry is obtained using a performance evaluation criteria, namely surface performance factor (i.e. Colburn factor to Fanning friction



(a)



(b)

Fig. 10. (a) Heat transfer coefficient – Reynolds number (b) pressure drop – Reynolds number for channels with different wings width.

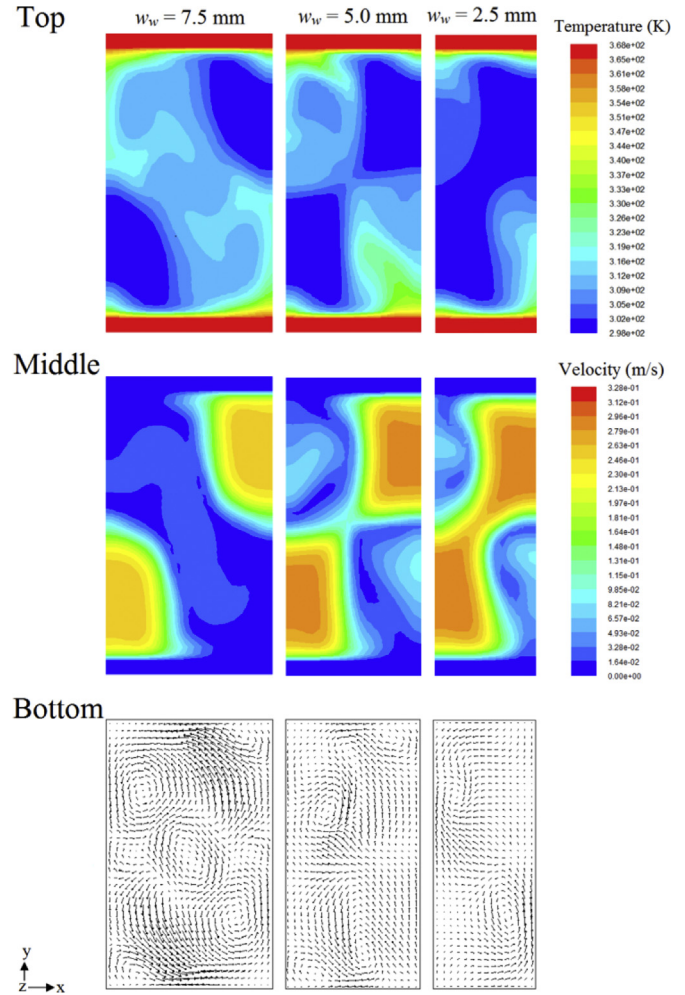


Fig. 11. (Top) Temperature contour (middle) velocity contour (bottom) stream line, i.e. secondary flows, on a normal surface for channels with different wings width at $Re = 1600$.

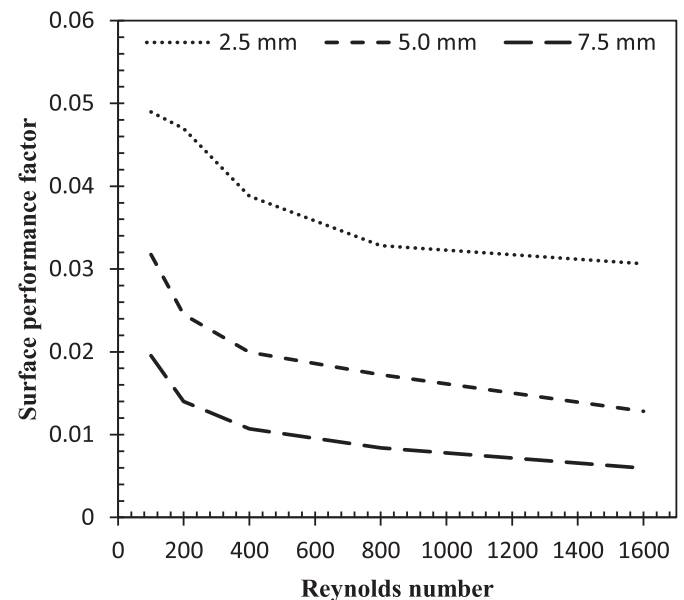


Fig. 12. Surface performance factor – Reynolds number for channels with different wing widths.

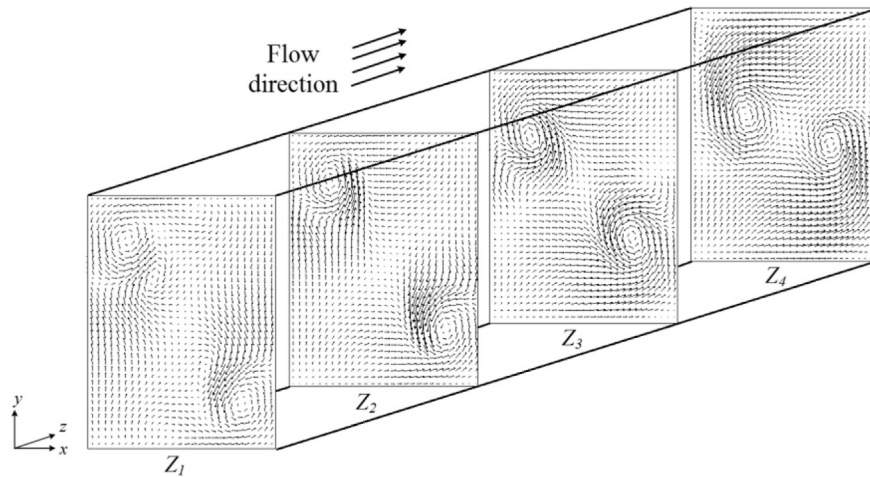


Fig. 13. Trend of secondary flows motion between two wings in a vortex-generator channel with dimension of ($w_h = w_w = 5.0$ mm, $L = 400$ mm, $l_p = 20$ mm, $t_p = 5.0$ mm, and $\alpha = \beta = 90^\circ$) at $Re = 1600$.

factor – j/f ratio [40]), Eq. (16), which are plotted versus Reynolds number.

$$\frac{j}{f} = \frac{Nu/RePr^{1/3}}{f} \quad (16)$$

For the better evaluation, an understanding of the flow behavior inside these channels is essential. Therefore, a good insight into the flow physics is obtained by analyzing the velocity field and stream lines in the fluid around the wings. In addition, to present detailed information about the locations of high heat transfer rates, the temperature distribution in the fluid is presented. For brevity, the flow and thermal field patterns are presented only for water.

5.1. Influence of wings shape

The heat transfer coefficient and pressure drop values against Reynolds number of the water flow inside the vortex-generator

channel with different wing shapes are presented in Fig. 5. The results show that the channel with the rectangular wings has the highest values, and the channels with the trapezoid and triangle wings come in the second and third. As depicted in Fig. 6, the high values of heat transfer coefficient are attributed to the high temperature gradient around the wings for the channel with the rectangular wings. Moreover, in the same geometrical parameters, the channel with rectangular wings provides more heat transfer area. Considering the maximum heat transfer performance as the main criterion, a parametric study is carried out for the vortex-generator channel with the rectangular wings in the next subsection.

5.2. Influence of geometrical parameters

The effects of wing height variations on the heat transfer coefficient and pressure drop values are presented in Fig. 7(a) and (b). The potential of heat transfer enhancement in a vortex-generator

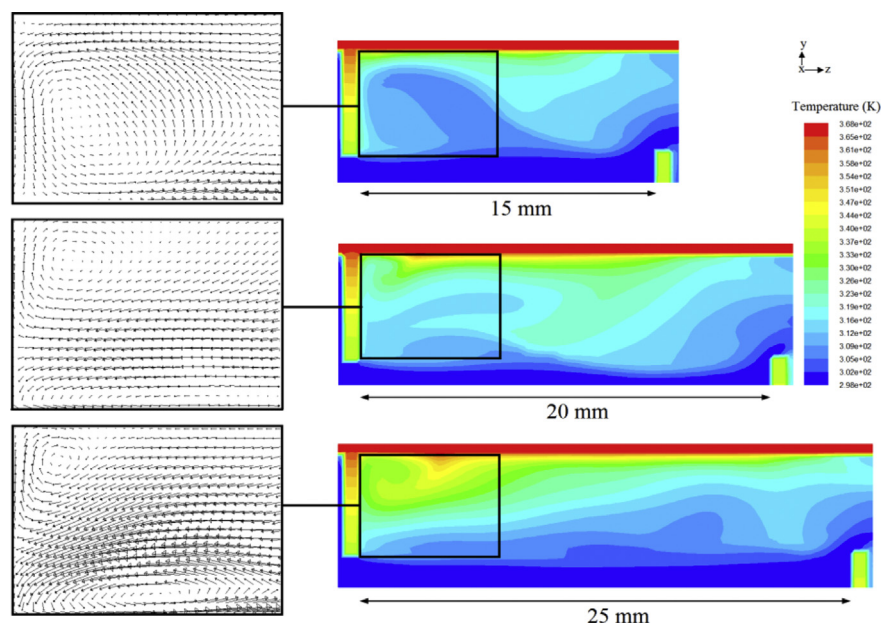


Fig. 14. Effect of longitudinal wings pitch variation on longitudinal vortex and temperature distribution between two wings.

channel depends on the location and strength of generated swirl flows (i.e. vortices) relative to the wings and hot walls. In the fact, in addition to Reynolds number, the variation of geometrical parameters affects the location and strength of longitudinal, transverse, and normal swirl flows. The parts of generated longitudinal and transverse vortices in the channels with different wing heights are illustrated in Fig. 8(a) and (b). In the channel with the lowest wing height (i.e. 2.5 mm), the expansion of main longitudinal vortex is considerable in comparison with the wing height. This causes a significant exchange between the core and the wall fluids. As the wings height increases, this ratio decreases, and a corner longitudinal vortex starts to generate. Due to the low temperature gradient in the location of generated corner vortex and also high distance between the main and the corner vortices, the effect of corner vortex on the heat transfer enhancement is not significant.

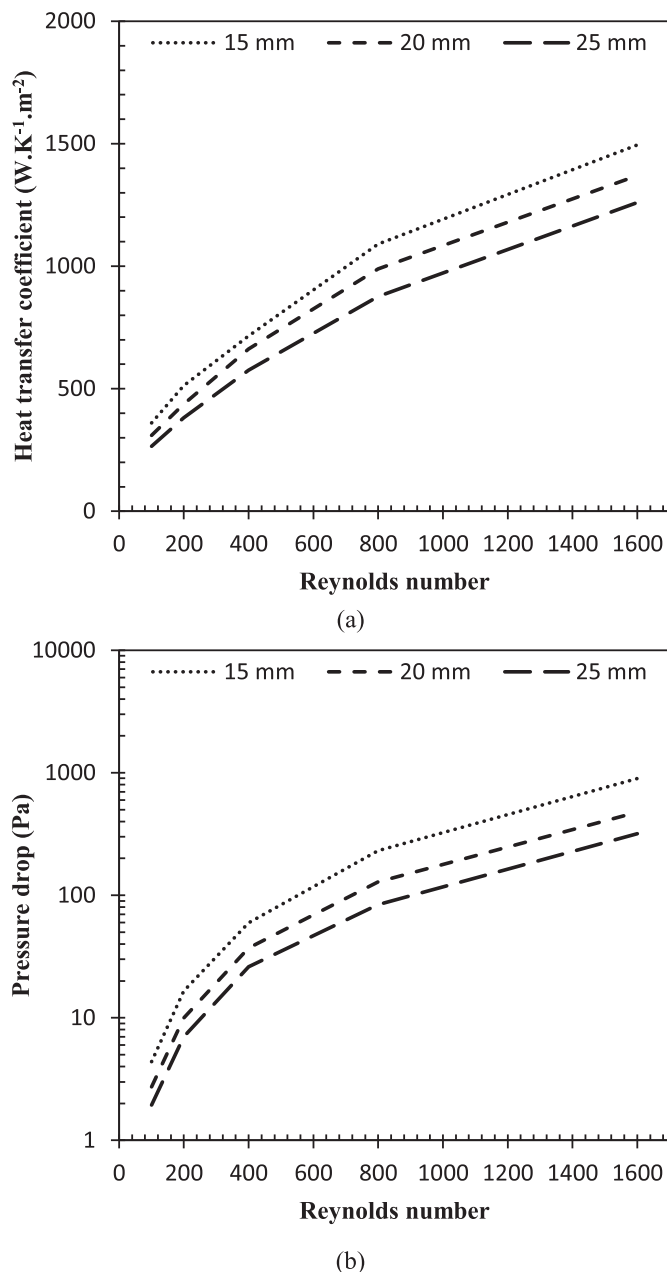


Fig. 15. (a) Heat transfer coefficient – Reynolds number (b) pressure drop – Reynolds number for channels with different longitudinal wing pitches.

Also, for the channel with higher wings height, the location of main longitudinal and transverse vortices recedes from the wing and wall, respectively. The mentioned mechanisms cause that the channels with lower wings height have the higher values of heat transfer coefficient.

To evaluate the performance of vortex-generator channel with different wing heights, the surface performance factor values versus Reynolds number are presented in Fig. 9. The results show that this factor decreases with Reynolds number, and the channel with the lowest wing height has the highest values.

Fig. 10(a) and (b) displays the variations of heat transfer coefficient and pressure drop values with Reynolds number for the channels with different wing widths. In contrast with the results of wings height, heat transfer coefficient and pressure drop enhance with increasing the wings width. The increase of heat transfer area is the main reason for the augmentation of the heat transfer rate. Furthermore, as illustrated in Fig. 11, the increase of channel width enhances the number and strength of normal vortices (i.e. secondary flows) and causes a heavy fluid exchange in the channel at the same Reynolds number. The secondary flows transfer the hot fluid from the near wall to the core section and increase the temperature gradient and heat transfer rate in the channel. The results show that at the selected surface, the channel with wings width of 2.5 mm has two induced vortices, the channel with wings width of 5.0 mm has two induced and two corner vortices, and the channel with wings width of 7.5 mm has two induced, two corner, and a main vortices.

Also, the surface performance factor variations with Reynolds number are shown in Fig. 12. While the channels with higher wings width have the greater values of heat transfer coefficient, the results show that increasing the wings width decreases the surface performance factor of vortex-generator channel. A comparison between Figs. 9 and 12 shows that the effect of wings width on the performance of vortex-generator channel is higher than that of the wings height.

In the same channel length, decreasing the longitudinal wings pitch not only increases the number of wings and thereby swirl flows in the channel, but also decreases the distance between two secondary flows (see Fig. 13). The longitudinal wings pitch

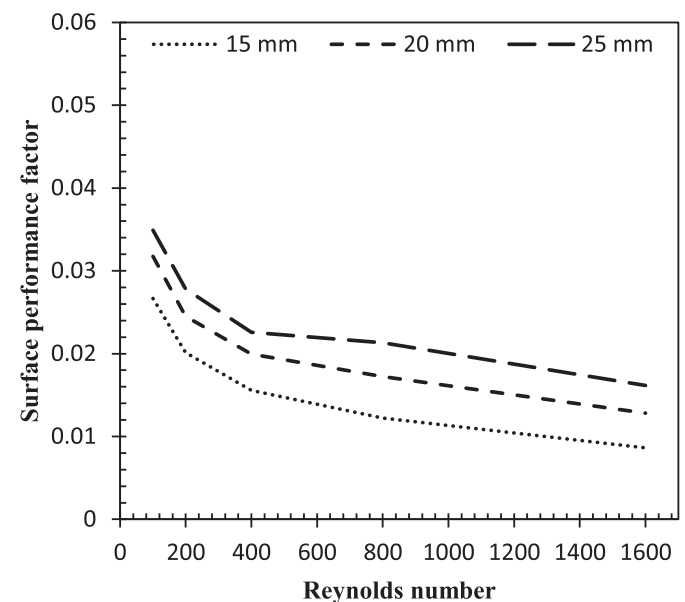


Fig. 16. Surface performance factor – Reynolds number for channels with different longitudinal wing pitches.

variations affects the longitudinal and transverse vortices in addition to the secondary flows. For instance, the effect of longitudinal wings pitch variations on the generated longitudinal vortex and temperature distribution between two wings are exhibited in Fig. 14. The results show that increasing the longitudinal distance reduces the strength of longitudinal vortex and temperature gradient in both the normal and the axial directions of channel.

The values of heat transfer coefficient and pressure drop for the channels with different longitudinal wing pitches are plotted against Reynolds number in Fig. 15(a) and (b). Based on the above descriptions, the channels with lower longitudinal wings pitch have the greater values of heat transfer coefficient and pressure drop. However, the net effect of this geometrical parameter

explains the increased surface performance factor for the higher longitudinal wing pitches (see Fig. 16).

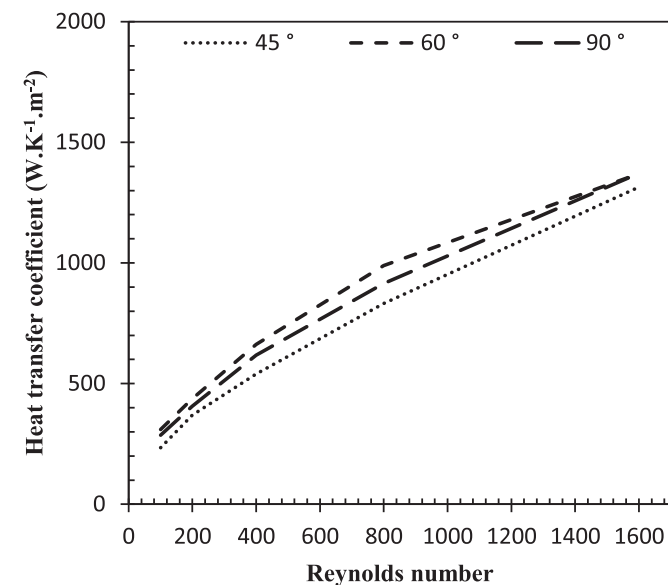
Similar to the longitudinal wings pitch, the heat transfer coefficient and pressure drop get higher values for the channels with the lower transverse wings pitch, while the greater values of the surface performance factor are obtained for the higher transverse wing pitches. It is interesting to note that in the studied range, the effect of transverse wings pitch on the surface performance factor is higher than that of the longitudinal wings pitch.

The influence of wings attach angle on the heat transfer coefficient and pressure drop values versus Reynolds number are plotted in Fig. 17(a) and (b). The results signify that the channel with wings attach angle of 60° has the highest values of heat transfer coefficient, whereas the highest values of pressure drop are obtained for the channel with wings attach angle of 90° . As presented in Fig. 18, the longitudinal vortex behind the wings with attach angle of 60° are more strength than that of 90° and 45° ones. However, as depicted in Fig. 19, the surface performance factor values of 45° is the highest, and that of 60° takes the second place, while that of 90° is the lowest. Also, the results show that the heat transfer coefficient and pressure drop values augment with increasing the attack angle. The channel with attack angle of 90° has the highest pressure drop due to blocking the fluid flow by the wings. However, this channel considerably increases the heat transfer coefficient. Furthermore, the surface performance factor gets the higher values for the lower attack angles (the performance factor of 45° is the highest at the studied range).

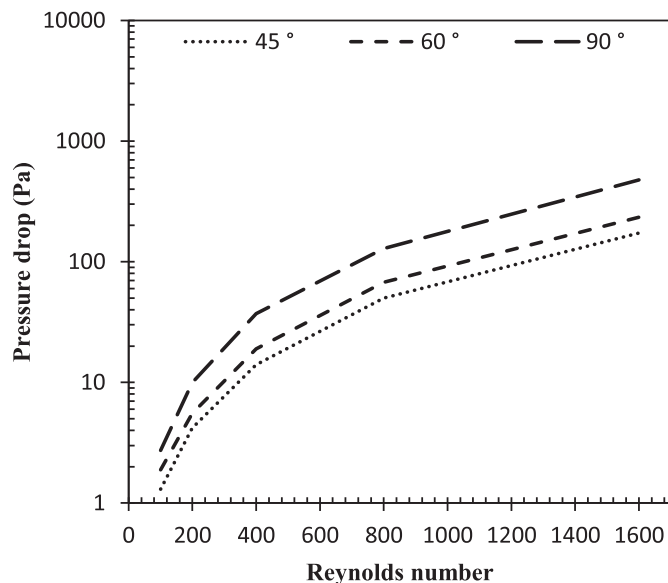
5.3. Nusselt number correlations

For easier engineer calculation, correlations were developed for Nusselt number by regression analysis coefficients on basis of 225 test data points from 15 models with differential geometrical parameters. The corresponding correlations for different working fluids are given as follows.

$$\text{Nu}_{\text{Water}} = \exp(-0.6678) \text{Re}^{0.5631} \left(\frac{w_w}{w_h} \right)^{0.1005} \times \left(\frac{l_p}{t_p} \right)^{0.0248} \left(\frac{\alpha}{\beta} \right)^{-0.1418} \quad (17)$$



(a)



(b)

Fig. 17. (a) Heat transfer coefficient – Reynolds number (b) pressure drop – Reynolds number for channels with different wing attach angles.

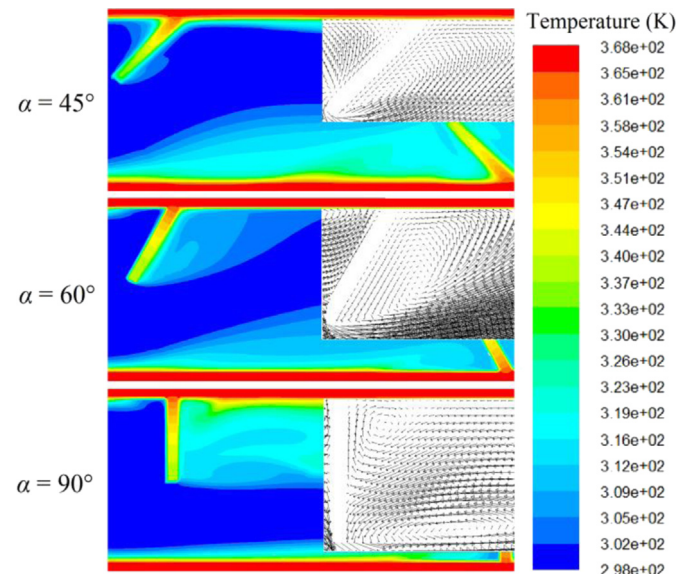


Fig. 18. Effect of wings attach angle on temperature gradient and longitudinal vortices behind wings.

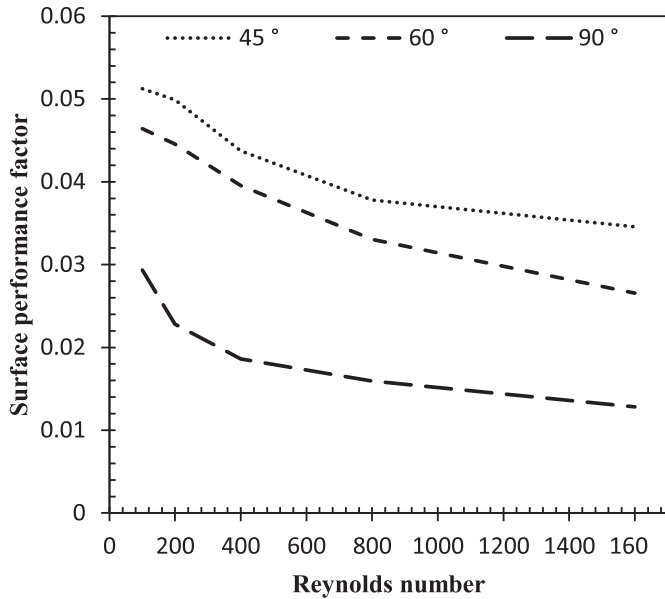


Fig. 19. Surface performance factor – Reynolds number for channels with different attach angles.

$$Nu_{Oil} = \exp(0.9102) Re^{0.4250} \left(\frac{w_w}{w_h} \right)^{0.2342} \left(\frac{l_p}{t_p} \right)^{0.0253} \left(\frac{\alpha}{\beta} \right)^{-0.0425} \quad (18)$$

$$Nu_{EG} = \exp(1.9694) Re^{0.2898} \left(\frac{w_w}{w_h} \right)^{0.4509} \left(\frac{l_p}{t_p} \right)^{0.0400} \left(\frac{\alpha}{\beta} \right)^{0.0713} \quad (19)$$

It can be concluded that the mean and maximum deviations between the presented Nusselt number correlations and the simulated results are 9.2% and 19.5% for water, 7.5% and 14.3% for oil, and 7.8% and 15.2% for ethylene glycol.

6. Conclusion

A parametric CFD study method was used to analyze the channel designed parameters affecting the performance of a plate-fin heat exchanger, when the proposed vortex-generator was applied as core surface. To validate the numerical results, experiments were carried out for the water flow inside a reference model by using a precise fabricated experimental setup. Effects of geometrical parameters such as the wings height, wings width, channel length, longitudinal wings pitch, transverse wings pitch, wings attach angle, and wings attack angle were examined. The results were compared based on their heat transfer coefficient and pressure drop along with a performance evaluation criterion, namely surface performance factor. The following conclusions can be made from the current study:

- I. The channel with the rectangular wings increases the heat transfer performance of the PFHE about 58.3% and 26.2% compared to the channels with the triangular and trapezoid wings, respectively.
- II. The heat transfer coefficient and pressured drop values enhance as the wings height, longitudinal wings pitch, and transverse wings pitch decrease and the wings width, channel length, and wings attack angle increase.
- III. The considered performance factor shows that the channel with lower wings height, wings width, wings attach angle,

and wings attack angle and higher longitudinal wings pitch and transverse wings pitch is the most attractive from the viewpoint of PFHE volume saving.

- IV. Among the studied design parameters, the wings height has the greatest effect on the performance evaluation criteria, i.e. j/f ratio, and the wings attach angle comes in the second.
- V. Correlations were developed to predict Nusselt number of the conventional working fluids flow (water, oil, and ethylene glycol) inside the PFHEs with vortex-generator channels. These correlations predict the present simulation data with the mean deviations lower than 10%.

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Nomenclature

A_c	minimum free flow area, m^2
A_t	total surface area in contact with working fluid, m^2
C_p	specific heat, $J\ kg^{-1}\ K^{-1}$
D_h	hydraulic diameter, m
h	effective heat transfer coefficient, $W\ m^{-2}\ K^{-1}$
L	channel length, m
l_p	longitudinal pitch, m
P	pressure, Pa
T	temperature, K
t_p	transverse pitch, m
w_h	wing height, m
w_w	wing width, m
ΔP	pressure drop, Pa

Greek symbols

ρ	density, $kg\ m^{-3}$
μ	dynamic viscosity, $kg\ m^{-1}\ s^{-1}$
κ	thermal conductivity, $W\ m^{-1}\ K^{-1}$
α	wing attach angle ($^\circ$)
β	wing attack angle ($^\circ$)

Subscripts

f	fluid
w_{all}	wall
s	solid
x	spanwise coordinate
y	normal coordinate
z	streamwise coordinate

Dimensionless groups

j	Colburn factor
f	Fanning friction factor
Nu	Nusselt number = hD_h/κ
Pr	Prandtl number = $\mu C_p/\kappa$
Re	Reynolds number = GD_h/μ

Acronyms

CFD	Computational Fluid Dynamics
EG	ethylene glycol
FVM	finite volume method
PFHE	plate-fin heat exchanger
SIMPLEC	semi implicit method for pressure linked equation
3-D	three-dimensional

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